s4_tilted_size_neutral

equity strategy | family: equity-neutralized | status: validated | derived from `rank_weighted`

What this strategy does

Trim 0.15, linear tilt, size-neutral per (quarter, side), priority-based cap trimming

Why it should work. Concentrate capital where the surprise is largest; neutralize size so it is not a hidden bet.

Design notes (from the strategy's source file).

Strategy 4 -- Trimmed, tilted, size-neutral, priority-capped.

Built after diagnosing why rank-weighting had the better Sharpe but the worse return: it traded ~4x more positions and saturated the liquidity/leverage caps. Strategy 4 drops the low-conviction middle of the SUE-rank distribution (trim 0.15), leans weight toward the extremes (tilt), keeps size neutrality, and when the leverage cap binds drops the LOWEST-conviction positions first instead of shaving everything equally.

parameter	value
base_unit_fraction	0.002
max_gross_leverage	1.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
beta_overlay	0.0

Performance

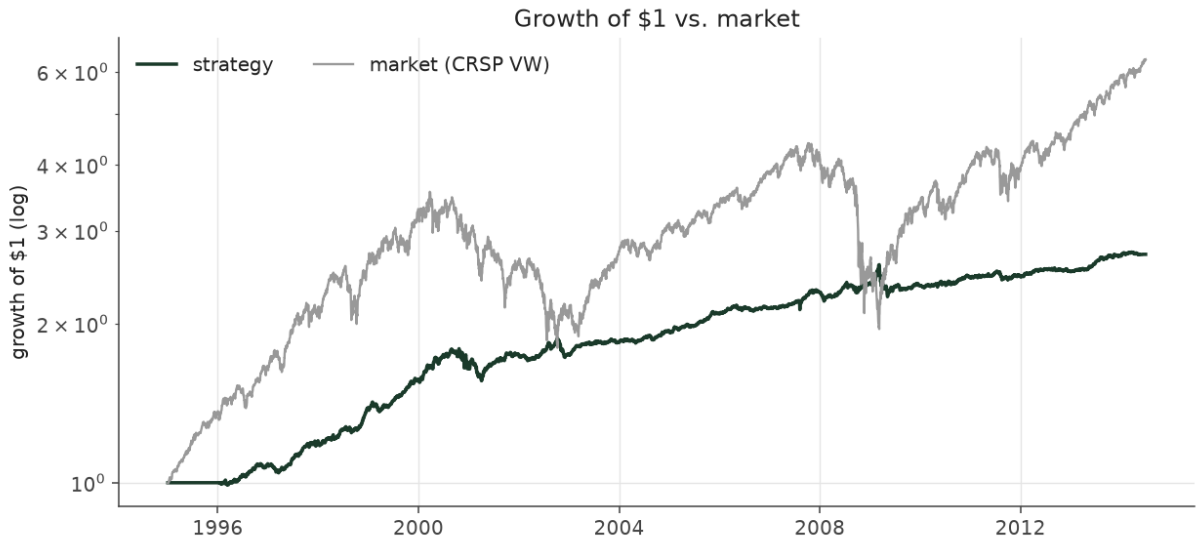
metric	s4 tilted size neutral	market (CRSP VW)
Annualized return	5.25%	9.94%
Annualized volatility	4.41%	19.35%
Sharpe (NAV-pct)	1.18	0.59
Sortino	1.70	
Max drawdown	-13.1%	-55.5%
Calmar	0.40	
Final NAV ($10M start)	$27,077,292	
Worst day	-2.43%	
% positive days	52.3%	
Sharpe 1996-2001	1.74	
Sharpe 2002-2007	1.31	
Sharpe 2008-2013	0.60	

Correlation of daily returns with the market: **0.05**.

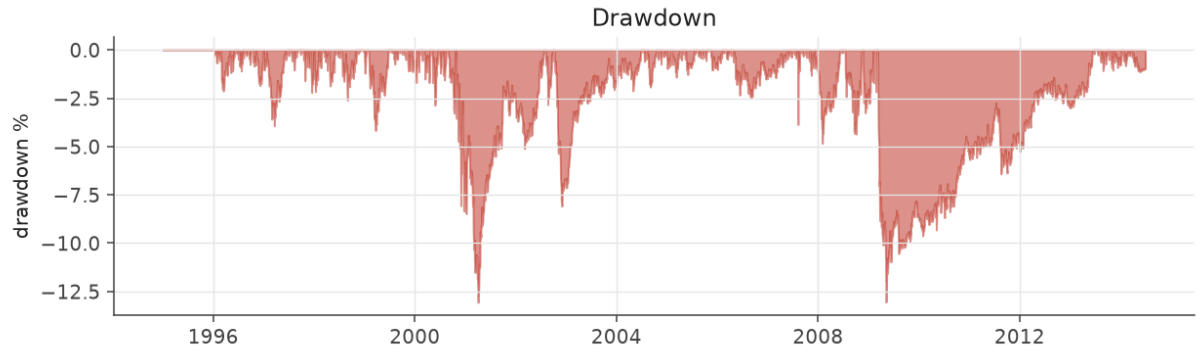
• best Sharpe among equity: `s10_vol_rank_tilt` (1.81); this one: 1.18

• best return among equity: `s6_beta_overlay_aggressive` (0.15); this one: 0.0525

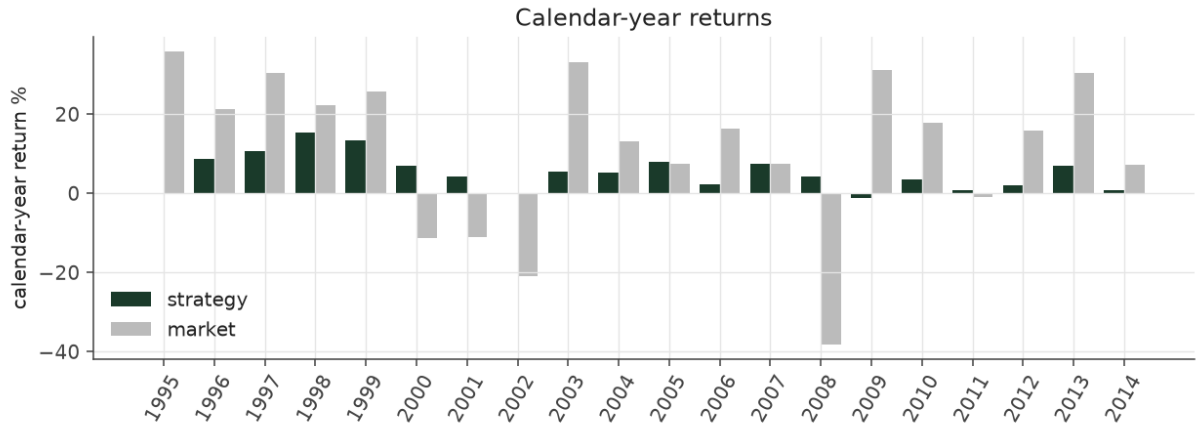
• best Calmar among equity: `s10_vol_rank_tilt` (0.614); this one: 0.402



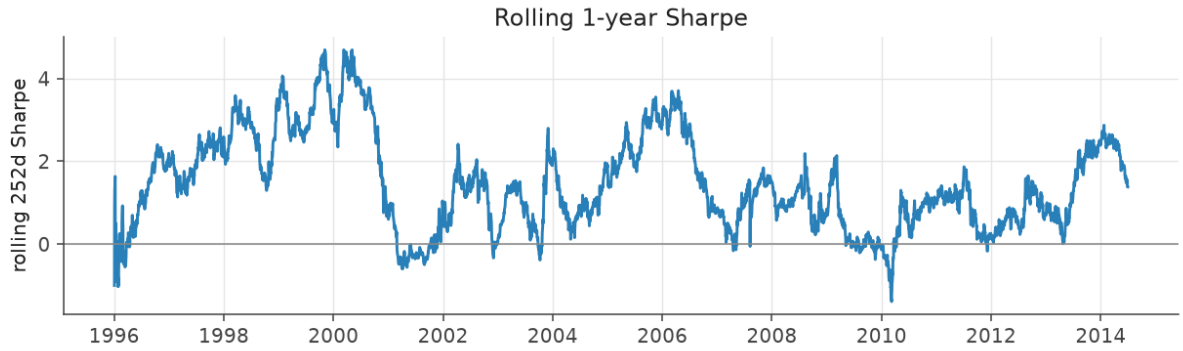
NAV (log scale)



Drawdown from running peak

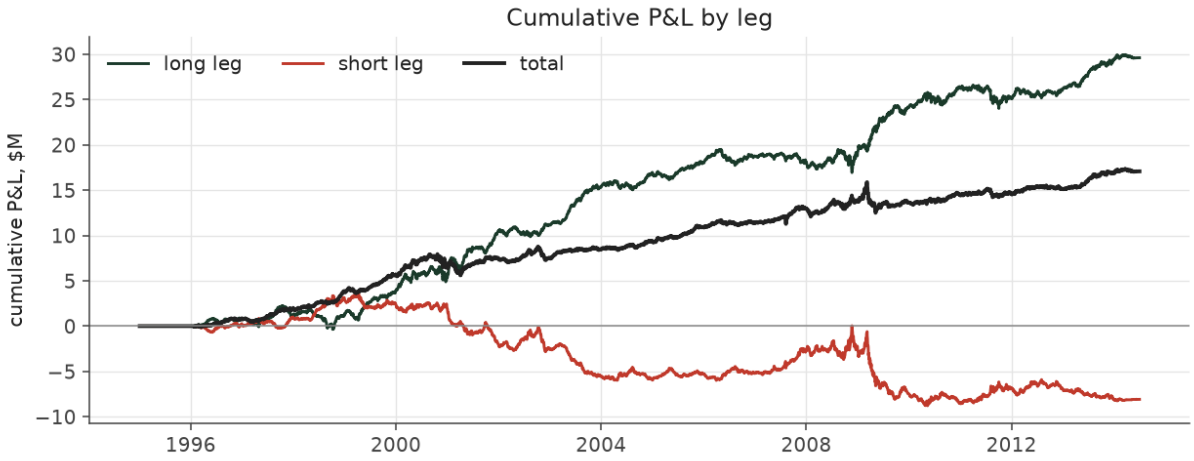


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	29.61	-8.08
P&L after costs ($M)	27.48	-10.41
of which common (average-event) component ($M)	16.38	-16.89
of which excess / PEAD component ($M)	13.23	8.81
excess minus costs ($M)	11.11	6.49
gross return per $ traded (bp)	297.54	-77.04
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	1.21	-0.29

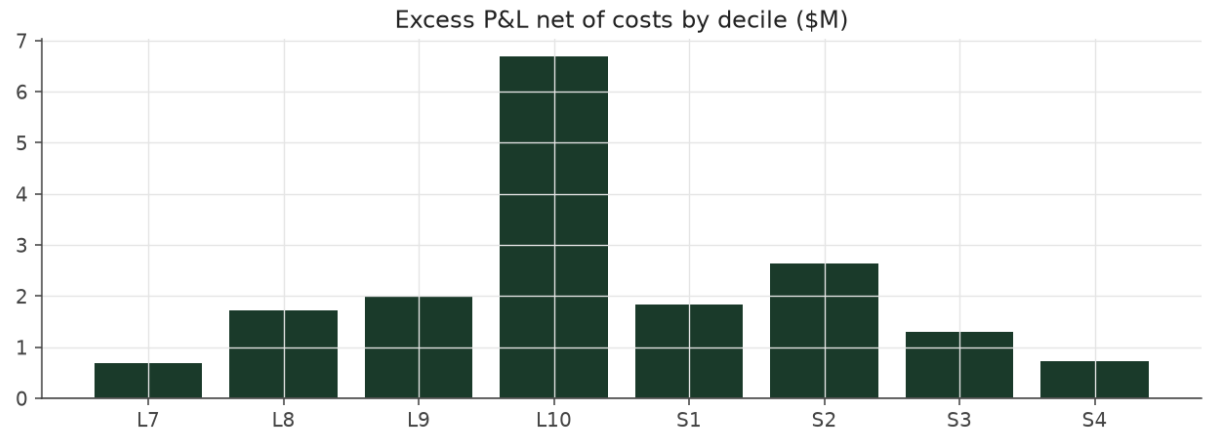
Daily P&L correlation between the two legs: **-0.78**.

• **The short leg loses money in absolute terms** (-8.1M before costs) while the long leg earns +29.6M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 16.4M / 16.9M. Stripping that common component, the surprise signal itself earned +13.2M on the long side and +8.8M on the short side.

• The two legs' daily P&L are -0.78 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.21 long / -0.29 short vs 1.18 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	5,109	54.41	2.00	0.82	0.13	0.69	3.9%	126.64
long	8	15,516	217.64	5.62	2.21	0.48	1.73	9.8%	79.41
long	9	17,587	322.33	6.68	2.67	0.67	2.00	11.4%	61.96
long	10	17,044	400.78	13.19	7.54	0.84	6.69	38.0%	166.97
short	1	16,800	412.85	-4.55	2.77	0.95	1.82	10.4%	44.21
short	2	17,012	330.93	-2.84	3.37	0.73	2.64	15.0%	79.86
short	3	15,637	230.86	-2.49	1.80	0.50	1.30	7.4%	56.41
short	4	7,653	74.51	-0.52	0.87	0.15	0.72	4.1%	96.34



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	12,635	277.38	4.88	6.46	1.07	5.39	30.6%	194.17
2	22,249	437.74	5.40	6.69	1.26	5.42	30.8%	123.88
3	24,545	445.39	3.82	4.73	0.93	3.79	21.6%	85.21
4	26,161	444.26	1.90	2.59	0.69	1.90	10.8%	42.68
5	26,768	439.55	1.08	1.58	0.48	1.09	6.2%	24.91

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	11,936	204.63	4.83	6.17	1.02	5.14	29.2%	251.41
2	22,441	467.78	4.88	6.85	1.40	5.44	30.9%	116.35
3	24,862	468.24	3.94	5.13	0.94	4.20	23.9%	89.64
4	26,282	458.63	2.03	2.52	0.64	1.88	10.7%	41.05

5	26,832	444.91	1.39	1.37	0.45	0.93	5.3%	20.88

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	42,328	715.87	-0.58	3.97	1.32	2.64	15.0%	36.93
41-60d	24,752	462.93	2.26	2.73	1.02	1.71	9.7%	36.92
61-90d	33,610	629.62	5.66	6.31	1.44	4.87	27.7%	77.40
>90d	11,668	235.90	9.74	9.04	0.67	8.37	47.6%	354.74

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	32,693	480.63	7.06	8.27	1.01	7.26	41.3%	151.11
2002-07	37,768	691.10	5.43	7.07	1.48	5.59	31.8%	80.85
2008-13	41,897	872.59	4.59	6.71	1.96	4.74	27.0%	54.37

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	21,073	387.69	3.73	3.00	0.85	2.15	12.2%	55.40
Chems	2,426	41.78	0.15	0.30	0.08	0.22	1.3%	53.56
Durbl	2,657	48.63	0.55	0.80	0.11	0.69	3.9%	142.34
Enrgy	5,779	101.46	1.40	1.75	0.20	1.55	8.8%	152.39
Hlth	10,938	200.75	2.46	2.43	0.48	1.95	11.1%	96.99
Manuf	11,395	202.91	1.11	2.05	0.43	1.62	9.2%	79.84
Money	16,527	302.05	2.08	3.88	0.64	3.23	18.4%	107.11
NoDur	4,812	85.02	0.71	0.85	0.17	0.68	3.9%	79.72
Other	17,162	317.77	2.40	3.37	0.75	2.62	14.9%	82.61
Shops	11,409	209.54	2.37	2.79	0.45	2.34	13.3%	111.48
Telcm	3,788	69.14	0.10	0.49	0.14	0.35	2.0%	51.12
Utils	4,392	77.58	0.01	0.33	0.14	0.19	1.1%	24.46

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	37,309	677.56	3.73	5.21	1.33	3.89	22.1%	57.36
low vol	37,771	683.62	6.69	8.71	1.64	7.07	40.2%	103.43
mid vol	35,726	647.23	6.25	7.66	1.41	6.25	35.5%	96.63

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	5.25%	0.00%
0.5%	1.1	5.03%	0.22%
1.0%	2.2	4.80%	0.45%
2.0%	4.3	4.31%	0.94%
5.0%	10.9	2.51%	2.73%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.